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Assessment of the ecological status of north-eastern Adriatic coastal waters (Istria, Croatia) using macroalgal assemblages for the European Union Water Framework Directive

LJILJANA IVEŠA*, DANIEL MARK LYONS and MASSIMO DEVESCOVI

Ruđer Bošković Institute, Center for Marine Research, G. Paliaga 5, 52210 Rovinj, Croatia

ABSTRACT

1. Based on the inclusion of macroalgae in the European Union Water Framework Directive as quality elements for the evaluation of the ecological status of coastal waters, the suitability of one (Ecological Evaluation Index, EEI) of several previously proposed evaluation methods in the particular ecological conditions of the northern Adriatic Sea was tested.

2. The EEI was assessed for 10 locations (polluted and putatively pristine) scattered along 60 km of the western Istrian coast. The sampling was performed seasonally at 1 and 3 m depth by destructive (determination of species cover and biomass) and non-destructive (determination of species coverage using digital photography) methods.

3. When assessed at 1 m depth the spatial scale weighted EEI for the west Istrian coast was 8.1, corresponding to an ecological status class (ESC) value of high. However, data for 3 m depth gave a spatial EEI of 6.72 which corresponds to an ESC value of good. Regressions of the ratio of ecological state group I (ESG I, i.e. thick leathery, calcareous and crustose species) over total algal abundance with the pollution gradient (obtained using principal components analysis (PCA) ordination of environmental variables) were significant at 3 m but not at 1 m depth. This was due to the high abundance of ESG I macroalgae *Corallina elongata* and *Cystoseira compressa* at 1 m depth at polluted stations. Similar regressions were obtained using cover, biomass and coverage.

4. It is concluded that the EEI method may be suitable for the classification of coastal waters in the northern Adriatic only in certain cases. A better assessment of ecological status using this method would require more realistic estimations based on the inclusion of data from several sampling depths. As all three abundance measures (cover, biomass, coverage) gave similar results, coverage (using digital photography) is suggested as being a preferred measure owing to the rapidity of sampling at several depths and less time-consuming laboratory work.

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KEY WORDS: Water Framework Directive; ecological evaluation index; benthic macroalgae; Istria; northern Adriatic; rocky bottom; *Cystoseira*; *Corallina*

INTRODUCTION

Trends in anthropogenic stress are reflected by the distribution patterns of marine species assemblages through alterations of biological processes and interactions among species (Dayton *et al*., 1995; Crowe *et al*., 2000; Archambault *et al*., 2001). Accordingly, marine organisms such as macroalgae, angiosperms, benthic invertebrate fauna and phytoplankton are included in the European Union Water Framework Directive (WFD) as quality elements for the evaluation of the ecological status of coastal and transitional waters and to act as a guidance tool for water quality management of coastal zones at local, national and international levels (EC, 2000). While macroinvertebrates living in soft-bottom benthic communities have traditionally been one of the sentinels of choice in water quality assessment (Dauer, 1993; Simboura and Zenetos, 2002), recent research has focused on developing indices for ecological status based on macrophytes.

Based on the requirements of the WFD, several methods have now been developed, each with its own advantages, to

*Correspondence to: L. Ivesa, Ruđer Bošković Institute, Center for Marine Research, G. Paliaga 5, 52210 Rovinj, Croatia. E-mail: ivesa@cim.irb.hr

assess the ecological status of transitional and coastal waters in various parts of Europe using assemblage. The CARLIT (cartography of littoral and upper-littoral rocky shore communities) method was developed and applied to coastal waters of the north-western Mediterranean (Ballesteros *et al.*, 2007; Mangialajo *et al.*, 2007). Sampling is non-destructive and consists of a survey of the entire coast with a small boat kept as close as possible to the shoreline to visually determine the type of growing macroalgal community. For coasts of the British Isles, Wells *et al.* (2007) proposed a metric scoring system using species richness, proportion of Chlorophyta, proportion of Rhodophyta, ratio of macroalgae subdivided in two ecological status groups (Wells, 2002), and proportion of opportunistic species as variables. The sampling was destructive and focused on the intertidal zone while the taxonomic determination was at the species level although a reduced species list was proposed. The monitoring of total and opportunistic algal cover in Danish coastal waters was based on a completely different sampling design. Samples were visually recorded by scuba along a depth gradient, usually to 12 m (Krause-Jensen *et al.*, 2007).

In the eastern Mediterranean, another index, termed the ecological evaluation index (EEI), was developed for categorizing both coastal and transitional waters of the Aegean coast (Orfanidis *et al.*, 2001, 2003). The method was based on destructive sampling of macroalgal assemblages at depths from 0 to 1 m, with the primary variable being macroalgal cover determined to at least the genus level. The concept of EEI is based on the macrophytobethos functional-morphological model of Littler and Littler (1980, 1984) where marine benthic macrophytes were divided into six functional-form groups: (A) Sheet-Group, (B) Filamentous-Group, (C) Coarsely Branched-Group, (D) Thick Leathery-Group, (E) Jointed Calcareous-Group and (F) Crustose-Group. Abundances of these groups are usually used as univariate variables in hypotheses testing factors affecting the composition of macroalgal assemblages (Benedetti-Cecchi *et al.*, 2001). Orfanidis *et al.* (2001) merged groups D, E and F into a single ecological state group (ESG I) and groups A, B and C into another (ESG II) thus creating the basis for the calculation of the EEI. Pristine environments are characterized by perennial species with low growth rates and reproductive potential and are grouped in ESG I. Highly stressed or disturbed marine environments are colonized by opportunistic species with high growth rates and reproductive potential and are grouped as members of ESG II. The evaluation of the EEI is performed by a cross comparison of the total cover of the two ESGs, resulting in a numerical scoring of locations which allows for spatial integration of the entire sampling area (Orfanidis *et al.*, 2001, 2003).

Herein, large-scale detailed surveys of macroalgal communities on the rocky north-eastern Adriatic coast have been performed to test the appropriateness of the EEI in this particular Mediterranean area. Locations influenced by urban pollution as well as locations that represent various pristine habitats along the shallow rocky Istrian shore were included in the study. Results obtained at two depths characterized by different types of macroalgal assemblages were compared. Subsequently, the response of the composition of macroalgal assemblages to urban pollution was studied using macroalgal abundance estimated by different methods (cover, biomass and coverage), including different methods of sampling (destructive and non-destructive).

METHODS

The study area

Sampling was carried out at 10 locations, during summer and autumn 2003 and winter and spring 2004, along 60 km of the western Istrian coast (northern Adriatic Sea, Croatia; Figure 1). All locations were characterized by a calcareous rocky substratum that was usually completely covered by macroalgal assemblages (100% coverage). Four locations were randomly selected in putatively pristine areas (locations 1 and 8–10). Two locations were situated on small islands at a distance of approximately 1 km or less from the town of Rovinj (locations 3 and 5) and three locations were randomly selected in the harbour areas of the town which were mainly subjected to urban pollution (locations 2, 4 and 6). Location 7 was in a non-urbanized area where the main sewage outlet of the town was located 1 km offshore at 35 m depth.

Environmental variables

Environmental variables were determined seasonally at the time of macroalgal sampling. Faecal coliform, streptococcal coliform and total coliform bacteria were determined by the membrane filter technique (APHA, 1980) in samples collected using sterile bottles at the sea surface. Nutrient concentrations were also determined in samples collected at the sea surface. Ammonium, nitrite and phosphate were determined according to Parsons *et al.* (1985). Chlorophyll *a*, oxygen saturation and transmission (measured as the percentage of light at 540 nm passing through 25 cm of seawater between a source and detector) were determined using a SeaBird SBE25 CTD profiler (Sea-Bird Electronics Inc.) that measured every 8 cm, from the surface to 5 m depth, and the data were averaged into 0.5 m bins.

Sampling and determination of macroalgae

At each location a permanent plot of approximately 10×10 m was marked using metal pegs hammered into the substratum for each of the depths 1 and 3 m on wide and structurally homogeneous, approximately horizontal rocky plateaus. At location 6 sampling at 3 m depth was not performed because of the lack of adequate rocky substratum. At each time of sampling, three 20×20 cm quadrats were haphazardly scattered over each permanent plot. This quadrat size is considered to be the representative minimal sampling area for infralittoral communities in the Mediterranean (Dhont and Coppejans, 1977; Boudouresque and Belsher, 1979). Quadrats were photographed using a digital underwater photo camera Olympus C-4040 Zoom (non-destructive sampling; Bianchi *et al.*, 2004). Then, macroalgae delimited by the quadrats were detached together with the adhering substratum using hammer and chisel and collected in polyethylene bags (destructive sampling; Bianchi *et al.*, 2004).

In the laboratory each sample was carefully sorted and macroalgae were identified at least to genus level and subdivided into ESG I and ESG II according to Orfanidis *et al.* (2001). The cover of each thallus, representing the surface covered in orthogonal projection, was determined according to Boudouresque (1971) and Cormaci *et al.* (2004). Results were expressed as percentage of total quadrat area (400 cm^2).

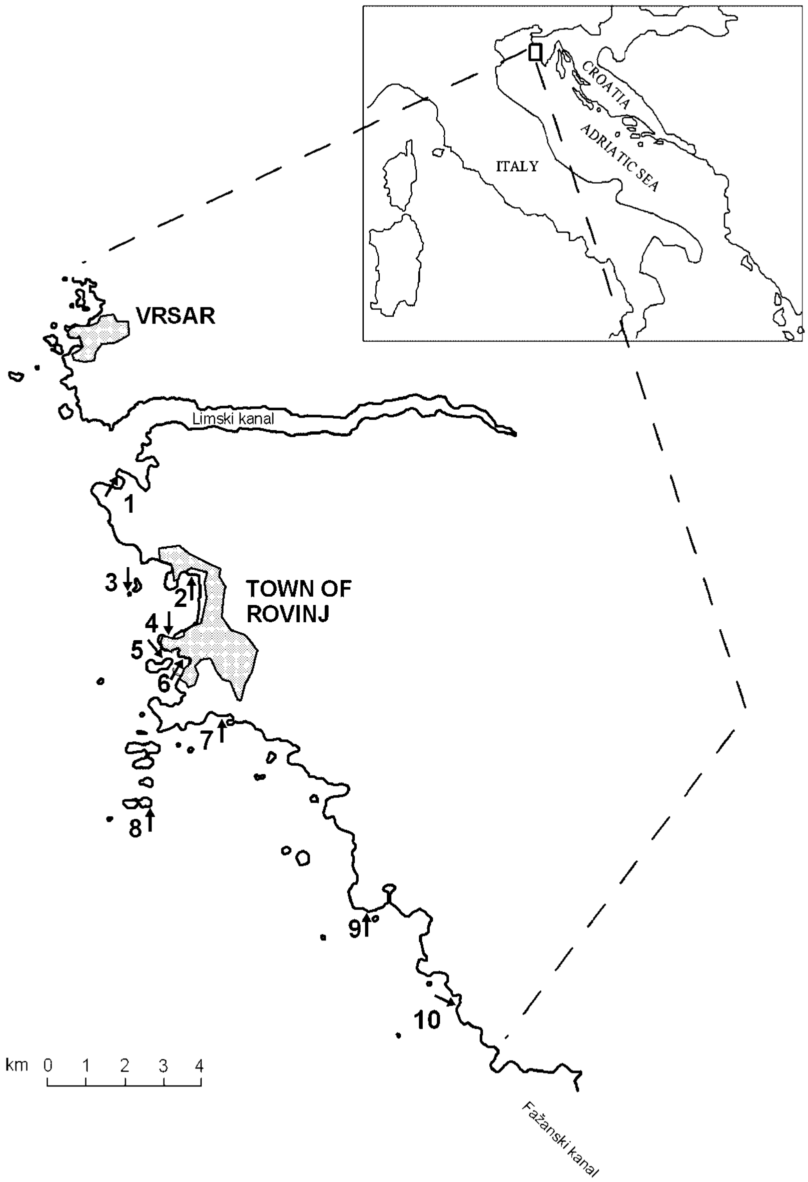

Figure 1. Map of the surveyed location along 60 km of the western Istrian coast. At location 7 the main sewage outlet of the town was located 1 km offshore at 35 m depth.

Additionally, thalli were thoroughly blotted free of adhering sea water using filter paper and the wet weight was measured using a Mettler Toledo PB 1502S balance. For further taxonomic identification, samples were conserved in 4% buffered formalin in sea water.

The area covered by individual genera visible in digital photographs was considered as the coverage. The total coverage by definition cannot exceed 100% while, because of macroalgal vertical stratification, total cover usually exceeds 100% (Bianchi *et al*., 2004). Coverage was expressed as the percentage of quadrat area covered by individual genera.

Assessment of the ecological evaluation index

The cover of ESG I and ESG II was calculated as pooled means of three replicates and four sampling dates. The cross comparison methodology to assess the EEI, the ecological status class (ESC) and spatial scale weighted EEI and ESC are given in Table 1 (Orfanidis *et al*., 2001, 2003). Spatial EEI represents the total of EEI values at each location multiplied by the respective portion of the coast (Orfanidis *et al*., 2003).

Data analysis

After four-root transformation and normalization, yearly means of environmental variables for each location were analysed by principal component analysis (PCA) using the software package PRIMER 6. To test relationships among macroalgal abundance estimated by different methods (cover, biomass, coverage) and the pollution level, data were transformed to a comparable form using the ratio of ESG I to the total of ESG I and ESG II (indicated as R_{ESG}):

$$R_{\mathrm{ESG}} = \frac{\mathrm{ESG\ I}}{\mathrm{ESG\ I} + \mathrm{ESG\ II}}$$

Table 1. Estimation of the ecological evaluation index (EEI) and the equivalent ecological status classes (ESC) using macroalgae grouped in two ecological status groups (ESG I and ESG II) according to Orfanidis *et al.* (2003)

Mean coverage of ESG I (%)	Mean coverage of ESG II (%)	ESC	EEI	Spatial scale weighted EEI and equivalent ESCs
0–30	0–30	Moderate	6	$\leqslant 6$ to >4 = Moderate
	> 30–60	Low	4	$\leqslant 4$ to >2 = Low
	> 60	Bad	2	2 = Bad
> 30–60	0–30	Good	8	$\leqslant 8$ to >6 = Good
	> 30–60	Moderate	6	$\leqslant 6$ to >4 = Moderate
	> 60	Low	4	$\leqslant 4$ to >2 = Low
> 60	0–30	High	10	$\leqslant 10$ to >8 = High
	> 30–60	Good	8	$\leqslant 8$ to >6 = Good
	> 60	Moderate	6	$\leqslant 6$ to >4 = Moderate

Table 2. Mean values of environmental variables assessed at inspected locations (L.) at four times from summer 2003 to spring 2004

L.	T. C.	F. C.	F. S.	NH_4^+	NO_2^-	PO_4^{3-}	Ch	$O_2\%$	Tr
1	37.75	0.00	2.50	0.83	0.50	0.04	0.48	105.50	74.75
2	61.50	3.75	18.25	1.49	0.61	0.16	0.55	103.75	62.59
3	56.00	0.25	1.25	0.34	0.49	0.04	0.44	104.75	60.23
4	930.00	475.00	479.25	0.70	0.79	0.14	0.54	102.50	71.97
5	50.00	2.00	9.75	0.42	0.62	0.04	0.46	103.25	47.31
6	746.50	71.25	481.50	0.66	0.53	0.09	0.47	105.75	57.50
7	37.25	1.25	9.00	0.40	0.70	0.04	0.42	102.25	74.95
8	0.50	0.00	2.00	0.39	0.63	0.03	0.48	104.25	74.87
9	1.00	0.00	0.75	0.31	0.52	0.03	0.35	105.00	76.71
10	1.00	0.00	0.25	0.61	0.63	0.06	0.30	103.00	63.01

Concentrations of total coliform (T. C.), faecal coliform (F. C.), and faecal streptococcal bacteria (F. S.) are given as number per 100 mL seawater. Concentrations of ammonia (NH_4^+), nitrite (NO_2^-), phosphate (PO_4^{3-}), and chlorophyll *a* (Ch) are in μmol L^{-1}. $O_2\%$ is the oxygen saturation. Tr is the transmission in %.

In this way data were transformed to proportions, ranging from 0 to 1, which are appropriate for analyses involving relationships among variables (Underwood, 1997). R_{ESG} values estimated for 1 and 3 m depth using macroalgal cover, biomass and coverage were regressed on PC1 scores of environmental variables for the inspected locations. Slopes and elevations of the regressions were compared according to Zar (1999).

RESULTS

Ordination of environmental variables

Yearly mean values of selected environmental variables are given in Table 2. The two-dimensional PCA ordination of these variables showed a strong pattern in relation to the urban pollution gradient (Figure 2). The first and second principal components accounted for 52% and 18% of the variability in the full matrix, respectively, thus providing an accurate summary of the sample relationships. The equation for PC1 was:

$$\begin{aligned} \mathrm{PC1} = & -0.439 \cdot \mathrm{FC} - 0.420 \cdot \mathrm{FS} - 0.413 \cdot \mathrm{TC} \\ & - 0.201 \cdot \mathrm{NH_4^+} - 0.250 \cdot \mathrm{NO_2^-} - 0.403 \cdot \mathrm{PO_4^{3-}} \\ & - 0.307 \cdot \mathrm{Ch} + 0.263 \cdot \mathrm{O_2\%} + 0.100 \cdot \mathrm{Tr} \end{aligned}$$

Coefficients of variables indicating urban pollution (faecal coliform = FC, faecal streptococcal = FS and total coliform bacteria = TC, ammonium = NH_4^+, nitrite = NO_2^-, phosphate = PO_4^{3-}, and chlorophyll *a* = Ch) were all negative. Coefficients of variables indicating pristine habitats (oxygen saturation = $O_2\%$, and transmission = Tr) were positive. Thus, PC1 represented an axis of decreasing urban pollution load. According to PCA ordination inspected locations could be subdivided into three categories: polluted (locations 2, 4 and 6), slightly polluted (locations 5 and 7) and pristine (locations 1, 3, 8, 9, and 10). Measurements in the water column showed an abrupt decrease in the urban pollution impact at a distance of approximately 1 km from the source (i.e. locations 3, 5 and 7).

The EEI at inspected locations

For the majority of locations, cross comparison of ESG I and ESG II total cover gave higher ESC at 1 m (Figure 3(a)) than at 3 m depth (Figure 3(b)). The ESC at location 6 was assessed only at 1 m depth because of the lack of rocky bottom at 3 m depth. Even though this station was polluted, the ESC was moderate. Only location 1 (pristine) and 7 (slightly polluted) showed the same ESC (good) at both depths. At 1 m depth, the ESC of polluted locations 2 and 4 was moderate and good, respectively. However, at 3 m depth the ESC of both locations was low, in accordance with the PCA ordination of environmental variables. High ESC values were found at locations 3 (pristine), 5 (slightly polluted), and 9 (pristine) at 1 m depth and good at location 3 but moderate at locations 5 and 9 at 3 m depth. Only at location 8 was the ESC lower at 1 m (moderate) than at 3 m (good) (Table 3).

These patterns were reflected by values of the spatial scale weighted EEI and ESC for the surveyed coast. The spatial EEI was 8.10 (ESC high) and 6.72 (ESC good) at 1 and 3 m depth, respectively (Table 3). This difference was mainly due to the

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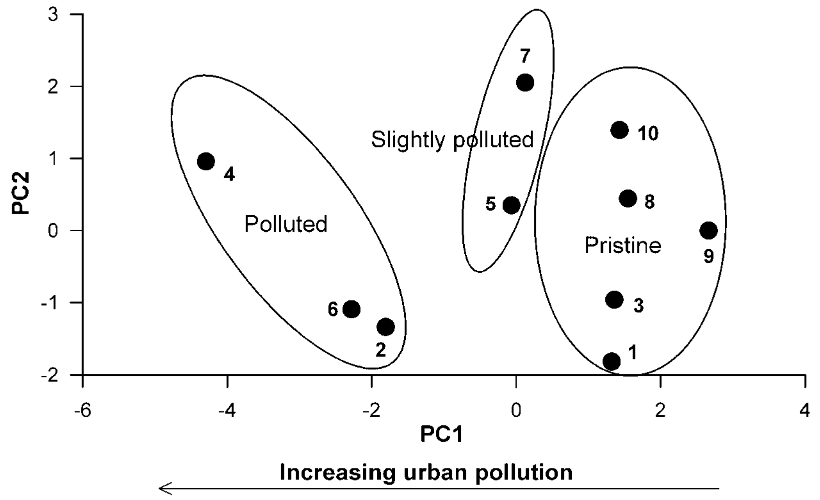

Figure 2. Two-dimensional PCA ordination of the nine environmental variables (4th-root transformed and normalized) for 10 locations along the western Istrian coast (variance explained: PC1 = 51.9%, PC2 = 18.0%).

high abundance at 1 m depth of only two macroalgal species of the ESG I in polluted stations: *Cystoseira compressa* (Esper) Gerloff and Nizamuddin f. *compressa* and *Corallina elongata* J. Ellis and Solander. At 1 m depth the brown alga *C. compressa* f. *compressa* was abundant at pristine locations 3 and 9, slightly polluted locations 5 and 7 and at polluted location 2 (Figure 4(a)). At pristine locations 1 and 10, two others *Cystoseira* species dominated at 1 m depth (Figure 4(b)): *Cystoseira barbata* (Stackhouse) C. Agardh v. *barbata* (location 1) and *Cystoseira crinita* Duby (location 10). The calcareous alga *C. elongata* dominated at polluted location 4 while the morphologically similar *Corallina officinalis* Linnaeus dominated at pristine location 8 (Figure 5).

At 3 m depth *C. compressa* f. *compressa* and *C. elongata* were no longer abundant. At this depth *C. barbata* v. *barbata* dominated at location 1 (Figure 4(b)) and the ESG I macroalgae *Padina pavonica* (Linnaeus) J. V. Lamouroux at locations from 7 to 10 (Figure 6(a)). Crustose macroalgae (ESG I) were particularly important only at pristine location 3. Among other locations (polluted or not) there was no clear difference in the dominance of crustose macroalgae (Figure 6(b)). At all locations the most abundant crustose macroalgae were *Neogoniolithon brassica-florida* (Harvey) Setchell and L. R. Mason and *Lithophyllum incrustans* Philippi.

Relationships between macroalgal composition and urban pollution

In general, regressions of R_{ESG}, estimated using cover, biomass or coverage, on PC1 scores of locations submitted to different levels of pollution revealed a linear trend. R_{ESG} increased with the decrease of urban pollution at both depths (Figures 7 and 8). At 1 m depth slopes were not statistically significant ($P = 0.120$, $P = 0.080$ and $P = 0.242$) for R_{ESG} calculated using cover, biomass and coverage, respectively.

However, at 3 m depth the cover, biomass and coverage R_{ESG} linearly, significantly increased ($P = 0.020$, $P = 0.031$, $P = 0.016$) using cover, biomass and coverage, respectively with the decrease of pollution (Figure 8). Slopes of these regressions were homogeneous ($F = 0.186$, $P > 0.05$ with 2 and 21 degrees of freedom (df)) as well as the elevations ($F = 1.714$, $P > 0.05$ with 2 and 23 df). Thus, at 3 m depth, relationships obtained using all three measures of macroalgal abundance could be considered identical.

DISCUSSION

In general, the composition of benthic macroalgal assemblages of the upper infralittoral (*sensu* Pérès and Gamulin-Brida, 1973) along the western Istrian coast reflects the intensity of urban pollution, which was evaluated using PCA ordination of environmental variables. However, calculated EEI values tended to change with depth, and were usually higher at 1 m than at 3 m depth, owing primarily to the abundance at 1 m depth of only two macroalgal species: *Corallina elongata* and *Cystoseira compressa* f. *compressa*. These two species are members of ESG I and are considered to be indicators of pristine habitats (Orfanidis *et al*., 2001). However, the presence of *Corallina elongata* does not seem to be unusual in polluted areas at lower depths (Soltan *et al*., 2001). Here, it could be that *C. elongatea,* as a coriaceous later-successional species, is adapted to withstand intense illumination and is physically able to endure wave shock and grazing as opposed to tender opportunistic species, which are more sensitive to these environmental factors (Littler and Littler, 1980). *C. compressa* f. *compressa* is a ubiquitous macroalga, which, along the east Adriatic coast, can be found in both exposed and sheltered, as well as moderately polluted and pristine habitats (Ercegović, 1952). In contrast, another methodology used in ecological quality assessment based on the cartography of littoral communities dominated by macroalgae does not group these two species with those highly sensitive to human disturbances (CARLIT, Ballesteros *et al*., 2007). For example, while the sensitivity level for *Cystoseira barbata* v. *barbata* and *Cystoseira crinita* is 20, that for *C. compressa* f. *compressa* is 12 and that for *C. elongata* is 8. Moreover, the sensitivity level for crustose algae such as *Neogoniolithon brassica-florida* and *Lithophyllum incrustans* is 6 while according to the EEI methodology these algae are all included in ESG I.

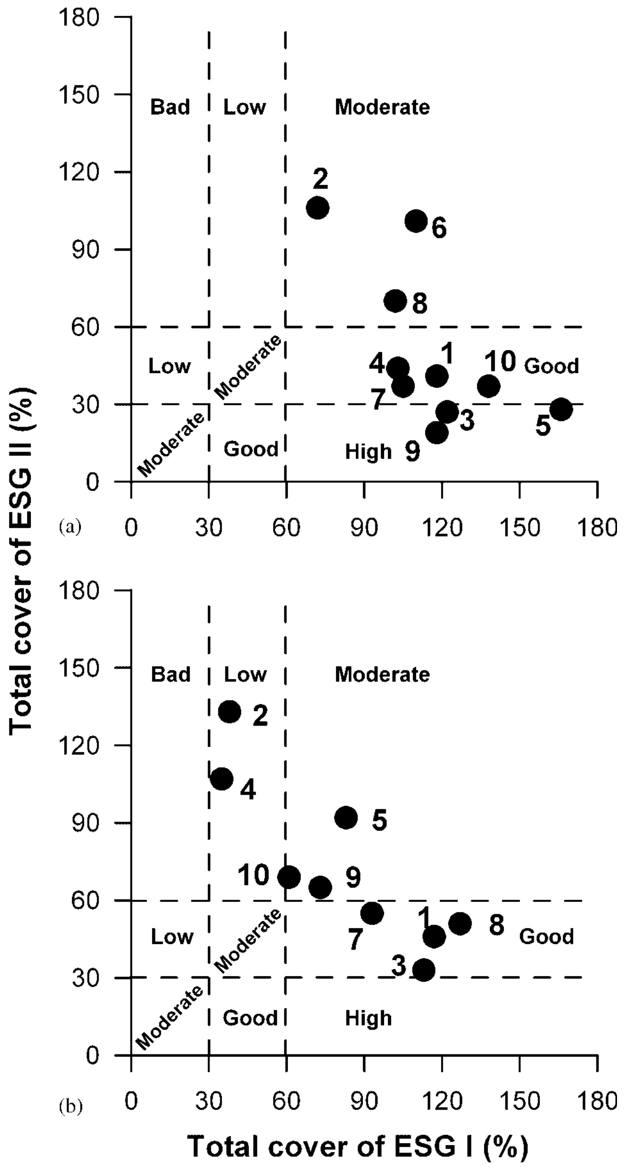

Figure 3. Categorized scatterplots of the macroalgal total cover (%) of ecological status groups (ESG I and ESG II) for locations along the western Istrian coast at 1 m (a) and 3 m depth (b). The vertical and horizontal lines divide the scatterplot in five ecological status classes.

The EEI methodology is based on the concept that the 'life-form' composition (i.e. incorporation of life history phenomena) of an algal community can be used to characterize the prevalent environmental conditions of a given habitat (Katada and Satomi, 1975; Littler and Littler, 1980, 1984). There is little general literature support for models grouping macroalgae by gross morphology, and many studies have shown that associated hypotheses are often incorrect (Padilla and Allen, 2000). However, morpho-functional grouping may be appropriate for large-scale monitoring surveys, addressing questions at the ecosystem level where there are too many species to consider them all individually. Based on this, it is difficult to see any clear advantage in changing the composition of species in the ESGs, for example, transposing *C. elongata* and *C. compressa* f. *compressa* from ESG I to ESG II. It is suggested that it may be more useful to assess ESGs at several locations and depths in each area that

Table 3. Ecological evaluation index (EEI) assessed at each location for two depths (1 and 3 m) and spatial EEI for 60 km of the west Istrian coast

Location	Portion of the coast	EEI (1 m)	EEI (3 m)	Spatial EEI (1 m)	Spatial EEI (3 m)
1	0.17	8	8	1.36	1.36
2	0.05	6	4	0.30	0.20
3	0.05	10	8	0.50	0.40
4	0.04	8	4	0.32	0.16
5	0.04	10	6	0.40	0.24
6	0.04	6	(4)[a]	0.24	(0.16)[a]
7	0.15	8	8	1.20	1.20
8	0.12	6	8	0.72	0.96
9	0.17	10	6	1.70	1.02
10	0.17	8	6	1.36	1.02
Spatial scale weighted EEI for the west Istrian coast:				8.10	6.72
Spatial scale weighted ecological status class:				High	Good

[a] Indicates that at location 6 the EEI was not assessed at 3 m depth, EEI = 4 (as for location 2 and 4) was assumed in spatial calculations.

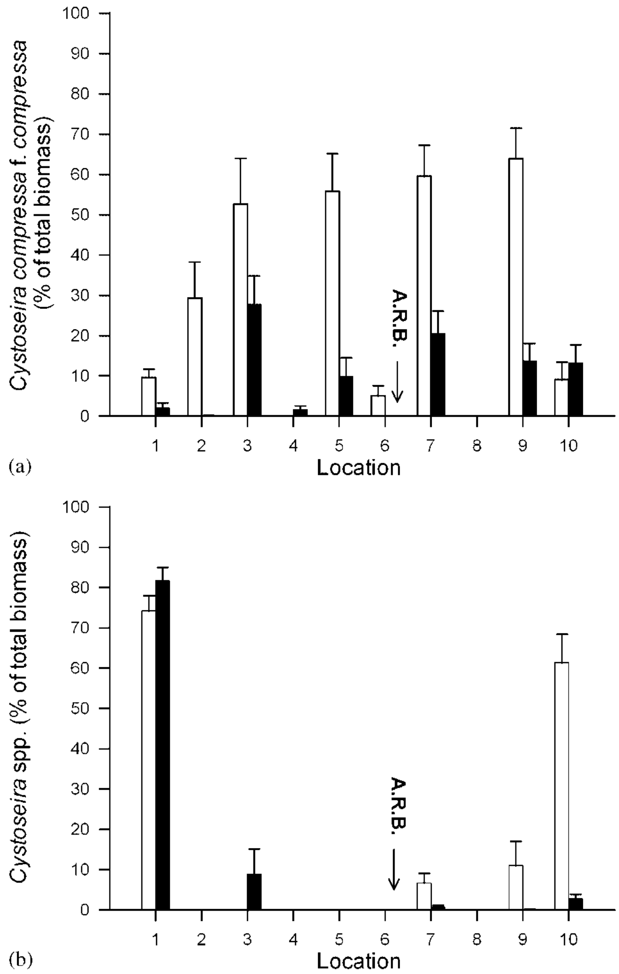

Figure 4. Abundance of *Cystoseira* species at inspected locations. (a) *Cystoseira compressa* v. *compressa*. (b) Other *Cystoseira* species pooled. *C. barbata* v. *barbata* and *C. crinita* dominated at locations 1 and 10, respectively. Empty bars = 1 m, full bars = 3 m depth. A. R. B. indicates the absence of rocky bottom at 3 m depth.

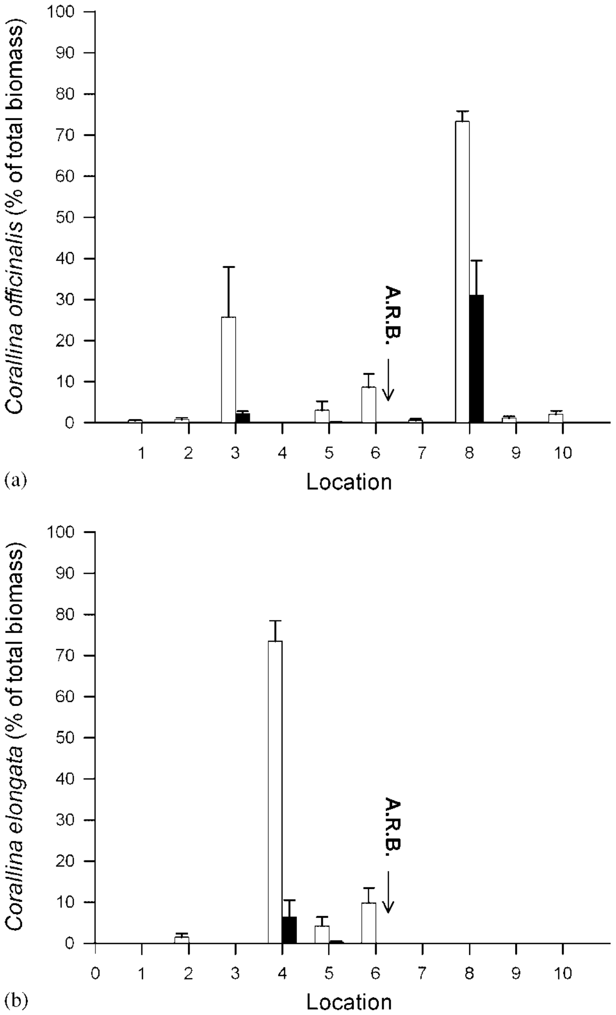

Figure 5. Abundance of *Corallina* species at inspected locations. (a) *C. officinalis*. (b) *C. elongata*. Empty bars = 1 m, full bars = 3 m depth. A. R. B. indicates the absence of rocky bottom at 3 m depth.

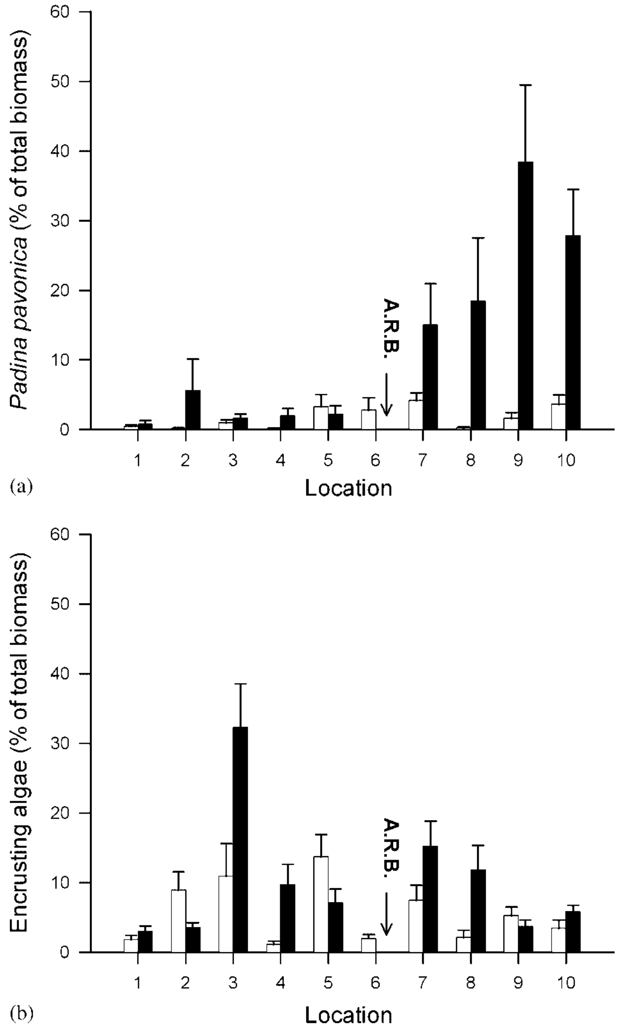

Figure 6. Abundance of *Padina pavonica* (a) and of pooled crustose algae (b) at the inspected locations. Empty bars = 1 m, full bars = 3 m depth. A. R. B. indicates the absence of rocky bottom at 3 m depth.

are believed to be exposed to similar environmental conditions and to test the inter-relationships between the ESGs abundance and the pollution intensity.

To test hypotheses about the relationship between the structure of macroalgal assemblages and the intensity of urban pollution we have introduced the ratio of ESG I to the total of ESG I and ESG II (R_{ESG}). While EEI values are categorical, R_{ESG} is a proportion and thus more useful for hypotheses testing. It may be calculated using different abundance measures such as cover, biomass and coverage, thus allowing comparison among them. The proportions of other macroalgal groups were also proposed as indices for the categorization of coastal waters of the British Isles (Wells *et al.*, 2007). Additionally, R_{ESG} is numerically similar to the ecological quality ratio (EQR; REFCOND, 2003) ranging from 0 (100% dominance of ESG II) to 1 (100% dominance of ESG I). R_{ESG} is not intended as a modification of the EEI, which is obtained by comparisons of macroalgal cover ranges. However, significant relationships between R_{ESG} and the pollution gradient could validate the grouping of ESGs which are at the base of the EEI classification method.

R_{ESG} significantly increased with the decrease of urban pollution at 3 m but not at 1 m depth. Consequently, for the northern Adriatic, it is suggested that it is preferable to assess the EEI at 3 m and not between 0 and 1 m as was originally proposed in the EEI methodology (Orfanidis *et al.*, 2001, 2003). Regressions of R_{ESG} calculated using cover, biomass (destructive sampling) and coverage (non-destructive sampling) showed consistent slopes and elevations suggesting that all these three measures of macroalgal abundance could be used in assessing the ecological status of coastal zones. Biomass wet weight has also been used by other authors to assess the impact of opportunistic species in transitional waters (Scanlan *et al.*, 2007). Biomass data are potentially important for the determination of the ecological status and can easily be determined in the laboratory in parallel with the estimation of cover (Scanlan *et al.*, 2007). Furthermore, Austoni *et al.* (2007) proposed the application of the specific exergy, i.e. a function of macroalgal assemblage biomass and its genetic information content, as an integrated index of environmental quality.

In spite of the significant relationship with the pollution gradient, R_{ESG} values obtained by coverage could be less than those estimated from cover and biomass. Reasons for this effect are twofold. Because of the vertical stratification of

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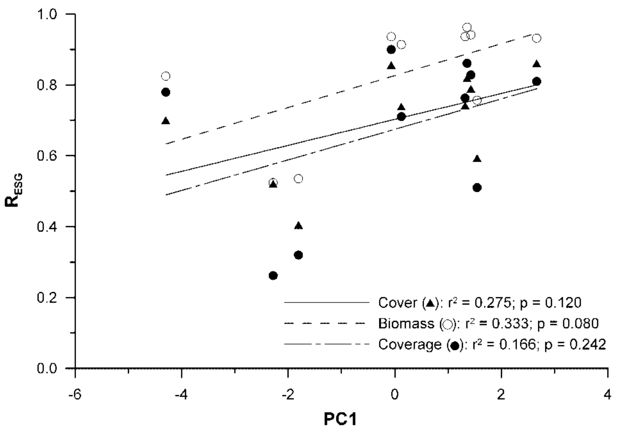

Figure 7. Linear regression of the ratio R_{ESG} = ESG I/(ESG I + ESG II) determined at 1 m depth using cover, biomass and coverage against the PC1 scores of environmental variables for inspected locations.

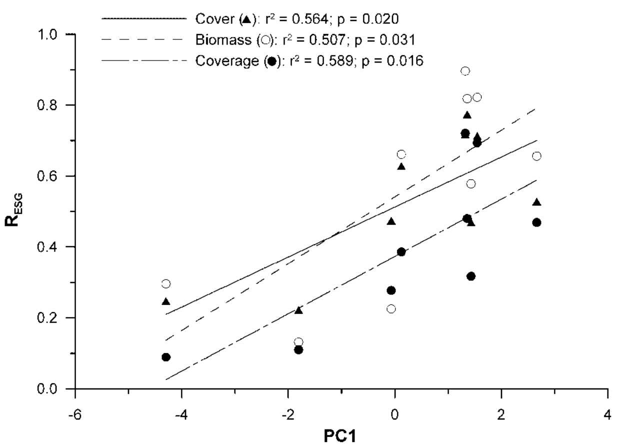

Figure 8. Linear regression of the ratio R_{ESG} = ESG I/(ESG I + ESG II) determined at 3 m depth using cover, biomass and coverage against the PC1 scores of environmental variables for inspected locations.

macroalgal assemblages, the under-storey Jointed Calcareous-Group and the Crustose-Group, which are members of ESG I, are not visible in digital photographs. Additionally, the coverage of epiphytes, as rule members of ESG II, may significantly influence the results. Non-destructive sampling allows the collection of a great number of replicates in a short time and is also the most appropriate method for work in natural reserves or in localities where the destruction of the rocky substratum is unacceptable. Therefore, to assess the appropriateness of the photograph sampling method for EEI classification further investigations are needed, for example considering under-storey assemblages after removal of upper canopies (Littler and Littler, 1985).

The results from this study are generally in agreement with those obtained for the western Mediterranean coast. For the depth range of 0.1 to 0.3 m the EEI methodology provided equivocal results at intermediate levels of nutrient concentration, i.e. approximately 0.5 $\mu mol\, L^{-1}$ phosphate and nitrite and from 1 to 3.5 $\mu mol\, L^{-1}$ ammonia (Arévalo *et al.*, 2007). However, using samples from 3 m depth stations with even lower nutrient concentration levels were successfully classified. Unlike urban pollution from a small town which may be detectable only within hundreds of metres from the sewage outfall (Arévalo *et al.*, 2007; this study), the EEI is intended as a tool to assess the ecological quality of large coastal areas where other disturbances may occur. For example, in the northern Adriatic Sea, rivers in the north-east of Italy are the most important source of nutrients and markedly influence eutrophication trends in this area (Degobbis *et al.*, 2000).

Historical data show that, in the northern Adriatic, eutrophication strongly influenced the spread of macroalgae of the genera *Cystoseira* and *Sargassum* and the brown alga *Fucus virsoides* J. Agardh (Munda, 2000). The first two genera belong to the ESG I group (Orfanidis *et al.*, 2001) while *F. virsoides* is a euryhaline species preferring sheltered habitats and is particularly abundant in the northern Adriatic; however, it does not occur in other Mediterranean areas. It belongs to the Thick Leathery-Group and thus may be considered an ESG I species. As *F. virsoides* colonizes the intertidal zone, in the future, its monitoring could be included in the CARLIT methodology for the specific case of the northern Adriatic.

According to the EEI classification method *Cystoseira* and *Sargassum* species have to be considered equivalent to species of both the Jointed Calcareous-Group and Crustose-Group. Because of eutrophication *Cystoseira* species could be eliminated from an area and replaced by species from the two aforementioned groups, without significant changes in EEI values. Such substitution has occurred in the past (Munda, 2000). After 1985, *Cystoseira* beds gradually recovered along the western Istrian coast (Babbini, 1991). Accordingly, dense beds of *Cystoseira barbata* v. *barbata* and *Cystoseira crinita* were recorded at stations 1 and 10, respectively. One of the reasons for this could be the decrease of phosphate concentration in north-east Italian rivers owing to the limitation of polyphosphate use in detergents (Tartari *et al.*, 1991). Therefore, given the presence of these *Cystoseira* species in the study area and the correlation of R_{ESG} at 3 m depth with environmental variables, the classification obtained by the EEI methodology could be sound. Regarding the usefulness of the methodology and its extension to other Adriatic and Mediterranean areas it must be specified that the methodology has some flaws. Major flaws of the EEI index are: (1) it changes with depth; (2) at 1 m depth it has no relation with water quality measured by standard methods; and (3) changes in the abundance of *Cystoseira* spp., reflecting eutrophication trends, are not explicitly recognized as important for the classification of coastal areas. Accordingly, further detailed analysis of macrophyte communities over a larger area at various depths and the testing of other proposed methodologies is required before definitive conclusions can be drawn on the most effective way for determining and monitoring the ecological status of waters of the northern Adriatic.

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